Output:

```
============================================================
> CONDITION: Calm Market
   Drift=0.00, Volatility=0.15
============================================================
  [PHYSICAL ACTION] LED: GREEN | Speed: 100% | Alarm: OFF
     -> OPTIMAL: Full market-making capacity engaged.
  Tick   0 | Mid: 94.50 | Spread: 0.57 | Inv: 0.50 | Equity: 10000.14
  [PHYSICAL ACTION] LED: GREEN | Speed: 100% | Alarm: OFF
     -> OPTIMAL: Full market-making capacity engaged.
  [PHYSICAL ACTION] LED: GREEN | Speed: 100% | Alarm: OFF
     -> OPTIMAL: Full market-making capacity engaged.
  Tick  20 | Mid: 117.50 | Spread: 0.82 | Inv: -0.50 | Equity: 10006.66
  [PHYSICAL ACTION] LED: GREEN | Speed: 100% | Alarm: OFF
     -> OPTIMAL: Full market-making capacity engaged.
  [PHYSICAL ACTION] LED: GREEN | Speed: 100% | Alarm: OFF
     -> OPTIMAL: Full market-making capacity engaged.
  Tick  40 | Mid: 95.47 | Spread: 0.86 | Inv: 0.50 | Equity: 10017.63
  [PHYSICAL ACTION] LED: GREEN | Speed: 100% | Alarm: OFF
     -> OPTIMAL: Full market-making capacity engaged.
```

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```
Tick  40 | Mid: 95.47 | Spread: 0.86 | Inv: 0.50 | Equity: 10017.63
[PHYSICAL ACTION] LED: GREEN | Speed: 100% | Alarm: OFF
   -> OPTIMAL: Full market-making capacity engaged.
[PHYSICAL ACTION] LED: GREEN | Speed: 100% | Alarm: OFF
   -> OPTIMAL: Full market-making capacity engaged.
Tick  60 | Mid: 95.18 | Spread: 0.84 | Inv: 1.50 | Equity: 10025.58
[PHYSICAL ACTION] LED: GREEN | Speed: 100% | Alarm: OFF
   -> OPTIMAL: Full market-making capacity engaged.
[PHYSICAL ACTION] LED: GREEN | Speed: 100% | Alarm: OFF
   -> OPTIMAL: Full market-making capacity engaged.
Tick  80 | Mid: 113.87 | Spread: 0.88 | Inv: -1.50 | Equity: 10051.91
[PHYSICAL ACTION] LED: GREEN | Speed: 100% | Alarm: OFF
   -> OPTIMAL: Full market-making capacity engaged.
[PHYSICAL ACTION] LED: GREEN | Speed: 100% | Alarm: OFF
   -> OPTIMAL: Full market-making capacity engaged.
Tick 100 | Mid: 120.38 | Spread: 0.85 | Inv: -3.00 | Equity: 9995.17
[PHYSICAL ACTION] LED: GREEN | Speed: 100% | Alarm: OFF
   -> OPTIMAL: Full market-making capacity engaged.
```

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```
   -> OPTIMAL: Full market-making capacity engaged.
[PHYSICAL ACTION] LED: GREEN | Speed: 100% | Alarm: OFF
   -> OPTIMAL: Full market-making capacity engaged.
Tick 120 | Mid: 123.92 | Spread: 0.88 | Inv: -5.50 | Equity: 9976.78
[PHYSICAL ACTION] LED: GREEN | Speed: 100% | Alarm: OFF
   -> OPTIMAL: Full market-making capacity engaged.
[PHYSICAL ACTION] LED: GREEN | Speed: 100% | Alarm: OFF
   -> OPTIMAL: Full market-making capacity engaged.
Tick 140 | Mid: 153.35 | Spread: 0.90 | Inv: -8.50 | Equity: 9842.96
[PHYSICAL ACTION] LED: GREEN | Speed: 100% | Alarm: OFF
   -> OPTIMAL: Full market-making capacity engaged.
[PHYSICAL ACTION] LED: GREEN | Speed: 100% | Alarm: OFF
   -> OPTIMAL: Full market-making capacity engaged.
Tick 160 | Mid: 123.07 | Spread: 0.85 | Inv: -13.00 | Equity: 10204.72
[PHYSICAL ACTION] LED: GREEN | Speed: 100% | Alarm: OFF
   -> OPTIMAL: Full market-making capacity engaged.
[PHYSICAL ACTION] LED: GREEN | Speed: 100% | Alarm: OFF
   -> OPTIMAL: Full market-making capacity engaged.
Tick 180 | Mid: 106.51 | Spread: 0.91 | Inv: -10.00 | Equity: 10473.58
```

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```
Tick 180 | Mid: 106.51 | Spread: 0.91 | Inv: -19.00 | Equity: 10473.58
[PHYSICAL ACTION] LED: YELLOW | Speed: 50% | Alarm: OFF
   -> CAUTION: Reducing position limits, tightening risk controls.
[PHYSICAL ACTION] LED: YELLOW | Speed: 50% | Alarm: OFF
   -> CAUTION: Reducing position limits, tightening risk controls.
Tick 200 | Mid: 93.27 | Spread: 0.84 | Inv: -22.00 | Equity: 10748.60
[PHYSICAL ACTION] LED: YELLOW | Speed: 50% | Alarm: OFF
   -> CAUTION: Reducing position limits, tightening risk controls.
[PHYSICAL ACTION] LED: YELLOW | Speed: 50% | Alarm: OFF
   -> CAUTION: Reducing position limits, tightening risk controls.
Tick 220 | Mid: 73.68 | Spread: 0.80 | Inv: -28.00 | Equity: 11250.70
[PHYSICAL ACTION] LED: YELLOW | Speed: 50% | Alarm: OFF
   -> CAUTION: Reducing position limits, tightening risk controls.
[PHYSICAL ACTION] LED: YELLOW | Speed: 50% | Alarm: OFF
   -> CAUTION: Reducing position limits, tightening risk controls.
Tick 240 | Mid: 63.33 | Spread: 0.85 | Inv: -32.50 | Equity: 11547.63

--- Final Metrics for Calm Market ---
Sharpe Ratio        : 1.6114
```

Scanned with OKEN Scanner

```
--- Final Metrics for Calm Market ---
  Sharpe Ratio        : 1.6114
  Max Drawdown        : 4.50%
  Total PnL           : $1475.72
  PnL per Trade       : $6.00
  Fill Rate           : 100.0%
  Final Inventory     : -35.00 units
  Total Trades        : 246

============================================================
> CONDITION: Trending Up
   Drift=0.40, Volatility=0.40
============================================================
  [PHYSICAL ACTION] LED: GREEN | Speed: 100% | Alarm: OFF
     -> OPTIMAL: Full market-making capacity engaged.
  Tick   0 | Mid: 87.91 | Spread: 0.57 | Inv: -0.50 | Equity: 10000.14
  [PHYSICAL ACTION] LED: GREEN | Speed: 100% | Alarm: OFF
     -> OPTIMAL: Full market-making capacity engaged.
  [PHYSICAL ACTION] LED: GREEN | Speed: 100% | Alarm: OFF
```

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```
   -> OPTIMAL: Full market-making capacity engaged.
[PHYSICAL ACTION] LED: GREEN | Speed: 100% | Alarm: OFF
   -> OPTIMAL: Full market-making capacity engaged.
Tick  20 | Mid: 303.73 | Spread: 1.36 | Inv: 1.00 | Equity: 10262.52
[PHYSICAL ACTION] LED: GREEN | Speed: 100% | Alarm: OFF
   -> OPTIMAL: Full market-making capacity engaged.
[PHYSICAL ACTION] LED: GREEN | Speed: 100% | Alarm: OFF
   -> OPTIMAL: Full market-making capacity engaged.
Tick  40 | Mid: 275.93 | Spread: 1.53 | Inv: 1.50 | Equity: 10232.83
[PHYSICAL ACTION] LED: GREEN | Speed: 100% | Alarm: OFF
   -> OPTIMAL: Full market-making capacity engaged.
[PHYSICAL ACTION] LED: GREEN | Speed: 100% | Alarm: OFF
   -> OPTIMAL: Full market-making capacity engaged.
Tick  60 | Mid: 626.75 | Spread: 1.43 | Inv: 2.00 | Equity: 11011.23
[PHYSICAL ACTION] LED: GREEN | Speed: 100% | Alarm: OFF
   -> OPTIMAL: Full market-making capacity engaged.
[PHYSICAL ACTION] LED: GREEN | Speed: 100% | Alarm: OFF
   -> OPTIMAL: Full market-making capacity engaged.
Tick  80 | Mid: 1234.52 | Spread: 1.49 | Inv: 2.50 | Equity: 12332.50
```

Scanned with OKEN Scanner

```
Tick  80 | Mid: 1234.52 | Spread: 1.49 | Inv: 2.50 | Equity: 12332.50
[PHYSICAL ACTION] LED: GREEN | Speed: 100% | Alarm: OFF
   -> OPTIMAL: Full market-making capacity engaged.
[PHYSICAL ACTION] LED: YELLOW | Speed: 50% | Alarm: OFF
   -> CAUTION: Reducing position limits, tightening risk controls.
Tick 100 | Mid: 2334.56 | Spread: 1.50 | Inv: 2.50 | Equity: 14863.81
[PHYSICAL ACTION] LED: GREEN | Speed: 100% | Alarm: OFF
   -> OPTIMAL: Full market-making capacity engaged.
[PHYSICAL ACTION] LED: YELLOW | Speed: 50% | Alarm: OFF
   -> CAUTION: Reducing position limits, tightening risk controls.
Tick 120 | Mid: 8026.91 | Spread: 1.37 | Inv: 4.00 | Equity: 30539.08
[PHYSICAL ACTION] LED: RED | Speed: 0% | Alarm: ON
   -> EMERGENCY STOP: Widening spreads to max, halting aggressive quoting.
[PHYSICAL ACTION] LED: GREEN | Speed: 100% | Alarm: OFF
   -> OPTIMAL: Full market-making capacity engaged.
Tick 140 | Mid: 12341.66 | Spread: 1.48 | Inv: 6.00 | Equity: 61048.26
[PHYSICAL ACTION] LED: GREEN | Speed: 100% | Alarm: OFF
   -> OPTIMAL: Full market-making capacity engaged.
```

Scanned with OKEN Scanner

```
   -> OPTIMAL: Full market-making capacity engaged.
[PHYSICAL ACTION] LED: YELLOW | Speed: 50% | Alarm: OFF
   -> CAUTION: Reducing position limits, tightening risk controls.
Tick 160 | Mid: 34959.07 | Spread: 1.21 | Inv: 10.50 | Equity: 236942.11
[PHYSICAL ACTION] LED: YELLOW | Speed: 50% | Alarm: OFF
   -> CAUTION: Reducing position limits, tightening risk controls.
[PHYSICAL ACTION] LED: RED | Speed: 0% | Alarm: ON
   -> EMERGENCY STOP: Widening spreads to max, halting aggressive quoting.
Tick 180 | Mid: 69063.18 | Spread: 1.37 | Inv: 14.00 | Equity: 621479.13
[PHYSICAL ACTION] LED: GREEN | Speed: 100% | Alarm: OFF
   -> OPTIMAL: Full market-making capacity engaged.
[PHYSICAL ACTION] LED: YELLOW | Speed: 50% | Alarm: OFF
   -> CAUTION: Reducing position limits, tightening risk controls.
Tick 200 | Mid: 351007.05 | Spread: 1.94 | Inv: 21.50 | Equity: 5553137.64
[PHYSICAL ACTION] LED: YELLOW | Speed: 50% | Alarm: OFF
   -> CAUTION: Reducing position limits, tightening risk controls.
[PHYSICAL ACTION] LED: YELLOW | Speed: 50% | Alarm: OFF
   -> CAUTION: Reducing position limits, tightening risk controls.
Tick 220 | Mid: 842365.25 | Spread: 1.40 | Inv: 28.50 | Equity: 18104137.53
```

Scanned with OKEN Scanner

```
 Tick 220 | Mid: 842365.25 | Spread: 1.40 | Inv: 28.50 | Equity: 18104137.53
 [PHYSICAL ACTION] LED: YELLOW | Speed: 50% | Alarm: OFF
    -> CAUTION: Reducing position limits, tightening risk controls.
 [PHYSICAL ACTION] LED: YELLOW | Speed: 50% | Alarm: OFF
    -> CAUTION: Reducing position limits, tightening risk controls.
 Tick 240 | Mid: 2476415.61 | Spread: 1.42 | Inv: 37.00 | Equity: 71564567.61

-- Final Metrics for Trending Up ---
 Sharpe Ratio        : 4.6939
 Max Drawdown        : 59.41%
 Total PnL           : $37731028.76
 PnL per Trade       : $209616.83
 Fill Rate           : 100.0%
 Final Inventory     : 40.00 units
 Total Trades        : 180

============================================================
 CONDITION: High Volatility
  Drift=0.00, Volatility=1.80
```

Scanned with OKEN Scanner

```
> CONDITION: High Volatility
  Drift=0.00, Volatility=1.80
============================================================
  [PHYSICAL ACTION] LED: GREEN | Speed: 100% | Alarm: OFF
     -> OPTIMAL: Full market-making capacity engaged.
  Tick   0 | Mid: 50.00 | Spread: 0.57 | Inv: 0.00 | Equity: 10000.29
  [PHYSICAL ACTION] LED: GREEN | Speed: 100% | Alarm: OFF
     -> OPTIMAL: Full market-making capacity engaged.
  [PHYSICAL ACTION] LED: GREEN | Speed: 100% | Alarm: OFF
     -> OPTIMAL: Full market-making capacity engaged.
  Tick  20 | Mid: 90.08 | Spread: 2.50 | Inv: 1.50 | Equity: 10104.83
  [PHYSICAL ACTION] LED: GREEN | Speed: 100% | Alarm: OFF
     -> OPTIMAL: Full market-making capacity engaged.
  [PHYSICAL ACTION] LED: GREEN | Speed: 100% | Alarm: OFF
     -> OPTIMAL: Full market-making capacity engaged.
  Tick  40 | Mid: 50.00 | Spread: 2.50 | Inv: 5.00 | Equity: 9997.58
  [PHYSICAL ACTION] LED: GREEN | Speed: 100% | Alarm: OFF
     -> OPTIMAL: Full market-making capacity engaged.
  [PHYSICAL ACTION] LED: GREEN | Speed: 100% | Alarm: OFF
```

Scanned with OKEN Scanner

```
   -> OPTIMAL: Full market-making capacity engaged.
[PHYSICAL ACTION] LED: GREEN | Speed: 100% | Alarm: OFF
   -> OPTIMAL: Full market-making capacity engaged.
Tick  60 | Mid: 59.09 | Spread: 2.50 | Inv: 6.50 | Equity: 10030.90
[PHYSICAL ACTION] LED: GREEN | Speed: 100% | Alarm: OFF
   -> OPTIMAL: Full market-making capacity engaged.
[PHYSICAL ACTION] LED: YELLOW | Speed: 50% | Alarm: OFF
   -> CAUTION: Reducing position limits, tightening risk controls.
Tick  80 | Mid: 167.36 | Spread: 2.50 | Inv: 10.00 | Equity: 10914.10
[PHYSICAL ACTION] LED: YELLOW | Speed: 50% | Alarm: OFF
   -> CAUTION: Reducing position limits, tightening risk controls.
[PHYSICAL ACTION] LED: RED | Speed: 0% | Alarm: ON
   -> EMERGENCY STOP: Widening spreads to max, halting aggressive quoting.
Tick 100 | Mid: 50.00 | Spread: 2.50 | Inv: 13.50 | Equity: 9489.93
[PHYSICAL ACTION] LED: RED | Speed: 0% | Alarm: ON
   -> EMERGENCY STOP: Widening spreads to max, halting aggressive quoting.
[PHYSICAL ACTION] LED: RED | Speed: 0% | Alarm: ON
   -> EMERGENCY STOP: Widening spreads to max, halting aggressive quoting.
Tick 120 | Mid: 94.30 | Spread: 2.50 | Inv: 20.50 | Equity: 10313.43
```

Scanned with OKEN Scanner

```
Tick 120 | Mid: 94.30 | Spread: 2.50 | Inv: 20.50 | Equity: 10313.43
[PHYSICAL ACTION] LED: RED | Speed: 0% | Alarm: ON
   -> EMERGENCY STOP: Widening spreads to max, halting aggressive quoting.
[PHYSICAL ACTION] LED: RED | Speed: 0% | Alarm: ON
   -> EMERGENCY STOP: Widening spreads to max, halting aggressive quoting.
Tick 140 | Mid: 50.00 | Spread: 2.50 | Inv: 28.50 | Equity: 9181.00
[PHYSICAL ACTION] LED: RED | Speed: 0% | Alarm: ON
   -> EMERGENCY STOP: Widening spreads to max, halting aggressive quoting.
[PHYSICAL ACTION] LED: RED | Speed: 0% | Alarm: ON
   -> EMERGENCY STOP: Widening spreads to max, halting aggressive quoting.
Tick 160 | Mid: 50.00 | Spread: 2.45 | Inv: 37.00 | Equity: 9118.41
[PHYSICAL ACTION] LED: RED | Speed: 0% | Alarm: ON
   -> EMERGENCY STOP: Widening spreads to max, halting aggressive quoting.
[PHYSICAL ACTION] LED: RED | Speed: 0% | Alarm: ON
   -> EMERGENCY STOP: Widening spreads to max, halting aggressive quoting.
Tick 180 | Mid: 119.28 | Spread: 2.50 | Inv: 44.50 | Equity: 11950.46
[PHYSICAL ACTION] LED: RED | Speed: 0% | Alarm: ON
   -> EMERGENCY STOP: Widening spreads to max, halting aggressive quoting.
[PHYSICAL ACTION] LED: RED | Speed: 0% | Alarm: ON
```

Scanned with OKEN Scanner

```
   -> EMERGENCY STOP: Widening spreads to max, halting aggressive quoting.
[PHYSICAL ACTION] LED: RED | Speed: 0% | Alarm: ON
   -> EMERGENCY STOP: Widening spreads to max, halting aggressive quoting.
Tick 200 | Mid: 78.17 | Spread: 2.50 | Inv: 52.00 | Equity: 9913.01
[PHYSICAL ACTION] LED: RED | Speed: 0% | Alarm: ON
   -> EMERGENCY STOP: Widening spreads to max, halting aggressive quoting.
[PHYSICAL ACTION] LED: RED | Speed: 0% | Alarm: ON
   -> EMERGENCY STOP: Widening spreads to max, halting aggressive quoting.
Tick 220 | Mid: 57.26 | Spread: 2.49 | Inv: 59.00 | Equity: 8821.64
[PHYSICAL ACTION] LED: RED | Speed: 0% | Alarm: ON
   -> EMERGENCY STOP: Widening spreads to max, halting aggressive quoting.
[PHYSICAL ACTION] LED: RED | Speed: 0% | Alarm: ON
   -> EMERGENCY STOP: Widening spreads to max, halting aggressive quoting.
Tick 240 | Mid: 50.00 | Spread: 2.50 | Inv: 68.00 | Equity: 6378.90

--- Final Metrics for High Volatility ---
 Sharpe Ratio        : 0.9372
 Max Drawdown        : 86.81%
 Total PnL           : $-3629.61
```

Scanned with OKEN Scanner

```
 Total PnL           : $-3629.61
 PnL per Trade       : $-18.81
 Fill Rate           : 100.0%
 Final Inventory     : 72.50 units
 Total Trades        : 193

============================================================
 CONDITION: Flash Crash
  Drift=-2.50, Volatility=2.50
============================================================
 [PHYSICAL ACTION] LED: GREEN | Speed: 100% | Alarm: OFF
    -> OPTIMAL: Full market-making capacity engaged.
 Tick   0 | Mid: 97.49 | Spread: 0.57 | Inv: 0.00 | Equity: 10000.29
 [PHYSICAL ACTION] LED: GREEN | Speed: 100% | Alarm: OFF
    -> OPTIMAL: Full market-making capacity engaged.
 [PHYSICAL ACTION] LED: GREEN | Speed: 100% | Alarm: OFF
    -> OPTIMAL: Full market-making capacity engaged.
 Tick  20 | Mid: 50.00 | Spread: 2.50 | Inv: 0.50 | Equity: 9931.32
```

Scanned with OKEN Scanner

```
Tick  20 | Mid: 50.00 | Spread: 2.50 | Inv: 0.50 | Equity: 9931.32
[PHYSICAL ACTION] LED: GREEN | Speed: 100% | Alarm: OFF
   -> OPTIMAL: Full market-making capacity engaged.
[PHYSICAL ACTION] LED: GREEN | Speed: 100% | Alarm: OFF
   -> OPTIMAL: Full market-making capacity engaged.
Tick  40 | Mid: 50.00 | Spread: 2.50 | Inv: 2.50 | Equity: 9851.14
[PHYSICAL ACTION] LED: GREEN | Speed: 100% | Alarm: OFF
   -> OPTIMAL: Full market-making capacity engaged.
[PHYSICAL ACTION] LED: GREEN | Speed: 100% | Alarm: OFF
   -> OPTIMAL: Full market-making capacity engaged.
Tick  60 | Mid: 50.00 | Spread: 1.57 | Inv: 5.50 | Equity: 9848.90
[PHYSICAL ACTION] LED: GREEN | Speed: 100% | Alarm: OFF
   -> OPTIMAL: Full market-making capacity engaged.
[PHYSICAL ACTION] LED: GREEN | Speed: 100% | Alarm: OFF
   -> OPTIMAL: Full market-making capacity engaged.
Tick  80 | Mid: 50.00 | Spread: 1.94 | Inv: 7.00 | Equity: 9844.08
[PHYSICAL ACTION] LED: GREEN | Speed: 100% | Alarm: OFF
   -> OPTIMAL: Full market-making capacity engaged.
[PHYSICAL ACTION] LED: YELLOW | Speed: 50% | Alarm: OFF
```

Scanned with OKEN Scanner

```
   -> OPTIMAL: Full market-making capacity engaged.
[PHYSICAL ACTION] LED: YELLOW | Speed: 50% | Alarm: OFF
   -> CAUTION: Reducing position limits, tightening risk controls.
Tick 100 | Mid: 50.00 | Spread: 2.50 | Inv: 7.50 | Equity: 9815.40
[PHYSICAL ACTION] LED: YELLOW | Speed: 50% | Alarm: OFF
   -> CAUTION: Reducing position limits, tightening risk controls.
[PHYSICAL ACTION] LED: YELLOW | Speed: 50% | Alarm: OFF
   -> CAUTION: Reducing position limits, tightening risk controls.
Tick 120 | Mid: 70.19 | Spread: 2.50 | Inv: 11.50 | Equity: 10028.80
[PHYSICAL ACTION] LED: YELLOW | Speed: 50% | Alarm: OFF
   -> CAUTION: Reducing position limits, tightening risk controls.
[PHYSICAL ACTION] LED: YELLOW | Speed: 50% | Alarm: OFF
   -> CAUTION: Reducing position limits, tightening risk controls.
Tick 140 | Mid: 50.00 | Spread: 2.13 | Inv: 12.00 | Equity: 9793.51
[PHYSICAL ACTION] LED: RED | Speed: 0% | Alarm: ON
   -> EMERGENCY STOP: Widening spreads to max, halting aggressive quoting.
[PHYSICAL ACTION] LED: RED | Speed: 0% | Alarm: ON
   -> EMERGENCY STOP: Widening spreads to max, halting aggressive quoting.
Tick 160 | Mid: 61.67 | Spread: 2.50 | Inv: 15.50 | Equity: 9797.50
```

Scanned with OKEN Scanner

```
 Tick 160 | Mid: 61.67 | Spread: 2.50 | Inv: 15.50 | Equity: 9797.50
 [PHYSICAL ACTION] LED: YELLOW | Speed: 50% | Alarm: OFF
    -> CAUTION: Reducing position limits, tightening risk controls.
 [PHYSICAL ACTION] LED: RED | Speed: 0% | Alarm: ON
    -> EMERGENCY STOP: Widening spreads to max, halting aggressive quoting.
 Tick 180 | Mid: 50.00 | Spread: 2.50 | Inv: 20.50 | Equity: 9428.24
 [PHYSICAL ACTION] LED: RED | Speed: 0% | Alarm: ON
    -> EMERGENCY STOP: Widening spreads to max, halting aggressive quoting.
 [PHYSICAL ACTION] LED: RED | Speed: 0% | Alarm: ON
    -> EMERGENCY STOP: Widening spreads to max, halting aggressive quoting.
 Tick 200 | Mid: 50.00 | Spread: 1.90 | Inv: 27.50 | Equity: 9414.18
 [PHYSICAL ACTION] LED: RED | Speed: 0% | Alarm: ON
    -> EMERGENCY STOP: Widening spreads to max, halting aggressive quoting.
 [PHYSICAL ACTION] LED: RED | Speed: 0% | Alarm: ON
    -> EMERGENCY STOP: Widening spreads to max, halting aggressive quoting.
 Tick 220 | Mid: 50.00 | Spread: 2.50 | Inv: 33.50 | Equity: 9292.34
 [PHYSICAL ACTION] LED: RED | Speed: 0% | Alarm: ON
    -> EMERGENCY STOP: Widening spreads to max, halting aggressive quoting.
 [PHYSICAL ACTION] LED: RED | Speed: 0% | Alarm: ON
```

Scanned with OKEN Scanner

```
 [PHYSICAL ACTION] LED: RED | Speed: 0% | Alarm: ON
    -> EMERGENCY STOP: Widening spreads to max, halting aggressive quoting.
 Tick 240 | Mid: 96.93 | Spread: 2.50 | Inv: 42.50 | Equity: 10876.30

-- Final Metrics for Flash Crash ---
 Sharpe Ratio        : 0.5183
 Max Drawdown        : 56.59%
 Total PnL           : $-1265.91
 PnL per Trade       : $-6.15
 Fill Rate           : 100.0%
 Final Inventory     : 46.00 units
 Total Trades        : 206

============================================================
 CONDITION: Mean-Reverting
  Drift=0.00, Volatility=0.90
============================================================
 [PHYSICAL ACTION] LED: GREEN | Speed: 100% | Alarm: OFF
```

Scanned with OKEN Scanner

```
[PHYSICAL ACTION] LED: GREEN | Speed: 100% | Alarm: OFF
   -> OPTIMAL: Full market-making capacity engaged.
Tick   0 | Mid: 140.87 | Spread: 0.57 | Inv: -0.50 | Equity: 10000.14
[PHYSICAL ACTION] LED: GREEN | Speed: 100% | Alarm: OFF
   -> OPTIMAL: Full market-making capacity engaged.
[PHYSICAL ACTION] LED: GREEN | Speed: 100% | Alarm: OFF
   -> OPTIMAL: Full market-making capacity engaged.
Tick  20 | Mid: 52.66 | Spread: 1.86 | Inv: -0.50 | Equity: 10029.79
[PHYSICAL ACTION] LED: GREEN | Speed: 100% | Alarm: OFF
   -> OPTIMAL: Full market-making capacity engaged.
[PHYSICAL ACTION] LED: GREEN | Speed: 100% | Alarm: OFF
   -> OPTIMAL: Full market-making capacity engaged.
Tick  40 | Mid: 219.48 | Spread: 2.50 | Inv: -1.50 | Equity: 9843.06
[PHYSICAL ACTION] LED: GREEN | Speed: 100% | Alarm: OFF
   -> OPTIMAL: Full market-making capacity engaged.
[PHYSICAL ACTION] LED: GREEN | Speed: 100% | Alarm: OFF
   -> OPTIMAL: Full market-making capacity engaged.
Tick  60 | Mid: 59.29 | Spread: 2.50 | Inv: -1.50 | Equity: 10059.14
[PHYSICAL ACTION] LED: GREEN | Speed: 100% | Alarm: OFF
```

Scanned with OKEN Scanner

```
Tick  60 | Mid: 59.29 | Spread: 2.50 | Inv: -1.50 | Equity: 10059.14
[PHYSICAL ACTION] LED: GREEN | Speed: 100% | Alarm: OFF
   -> OPTIMAL: Full market-making capacity engaged.
[PHYSICAL ACTION] LED: GREEN | Speed: 100% | Alarm: OFF
   -> OPTIMAL: Full market-making capacity engaged.
Tick  80 | Mid: 87.62 | Spread: 2.27 | Inv: -3.50 | Equity: 10009.00
[PHYSICAL ACTION] LED: GREEN | Speed: 100% | Alarm: OFF
   -> OPTIMAL: Full market-making capacity engaged.
[PHYSICAL ACTION] LED: GREEN | Speed: 100% | Alarm: OFF
   -> OPTIMAL: Full market-making capacity engaged.
Tick 100 | Mid: 57.04 | Spread: 2.10 | Inv: -5.50 | Equity: 10138.33
[PHYSICAL ACTION] LED: GREEN | Speed: 100% | Alarm: OFF
   -> OPTIMAL: Full market-making capacity engaged.
[PHYSICAL ACTION] LED: GREEN | Speed: 100% | Alarm: OFF
   -> OPTIMAL: Full market-making capacity engaged.
Tick 120 | Mid: 114.09 | Spread: 2.50 | Inv: -8.50 | Equity: 9849.96
[PHYSICAL ACTION] LED: GREEN | Speed: 100% | Alarm: OFF
   -> OPTIMAL: Full market-making capacity engaged.
[PHYSICAL ACTION] LED: RED | Speed: 0% | Alarm: ON
```

Scanned with OKEN Scanner

```
   -> OPTIMAL: Full market-making capacity engaged.
[PHYSICAL ACTION] LED: RED | Speed: 0% | Alarm: ON
   -> EMERGENCY STOP: Widening spreads to max, halting aggressive quoting.
Tick 140 | Mid: 414.28 | Spread: 2.50 | Inv: -12.50 | Equity: 6506.13
[PHYSICAL ACTION] LED: RED | Speed: 0% | Alarm: ON
   -> EMERGENCY STOP: Widening spreads to max, halting aggressive quoting.
[PHYSICAL ACTION] LED: GREEN | Speed: 100% | Alarm: OFF
   -> OPTIMAL: Full market-making capacity engaged.
Tick 160 | Mid: 188.78 | Spread: 1.94 | Inv: -19.50 | Equity: 9879.26
[PHYSICAL ACTION] LED: RED | Speed: 0% | Alarm: ON
   -> EMERGENCY STOP: Widening spreads to max, halting aggressive quoting.
[PHYSICAL ACTION] LED: RED | Speed: 0% | Alarm: ON
   -> EMERGENCY STOP: Widening spreads to max, halting aggressive quoting.
Tick 180 | Mid: 413.00 | Spread: 2.15 | Inv: -27.50 | Equity: 4588.45
[PHYSICAL ACTION] LED: YELLOW | Speed: 50% | Alarm: OFF
   -> CAUTION: Reducing position limits, tightening risk controls.
[PHYSICAL ACTION] LED: YELLOW | Speed: 50% | Alarm: OFF
   -> CAUTION: Reducing position limits, tightening risk controls.
Tick 200 | Mid: 50.00 | Spread: 2.27 | Inv: -36.00 | Equity: 15173.76
```

Scanned with OKEN Scanner

```
  Tick 200 | Mid: 50.00 | Spread: 2.27 | Inv: -36.00 | Equity: 15173.76
  [PHYSICAL ACTION] LED: YELLOW | Speed: 50% | Alarm: OFF
     -> CAUTION: Reducing position limits, tightening risk controls.
  [PHYSICAL ACTION] LED: RED | Speed: 0% | Alarm: ON
     -> EMERGENCY STOP: Widening spreads to max, halting aggressive quoting.
  Tick 220 | Mid: 161.33 | Spread: 2.42 | Inv: -44.00 | Equity: 10580.37
  [PHYSICAL ACTION] LED: RED | Speed: 0% | Alarm: ON
     -> EMERGENCY STOP: Widening spreads to max, halting aggressive quoting.
  [PHYSICAL ACTION] LED: RED | Speed: 0% | Alarm: ON
     -> EMERGENCY STOP: Widening spreads to max, halting aggressive quoting.
  Tick 240 | Mid: 65.81 | Spread: 2.50 | Inv: -52.00 | Equity: 14949.65

--- Final Metrics for Mean-Reverting ---
  Sharpe Ratio        : 1.0014
  Max Drawdown        : 55.56%
  Total PnL           : $5872.10
  PnL per Trade       : $39.95
  Fill Rate           : 100.0%
  Final Inventory     : -56.50 units
```

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```
  Sharpe Ratio        : 1.0014
  Max Drawdown        : 55.56%
  Total PnL           : $5872.10
  PnL per Trade       : $39.95
  Fill Rate           : 100.0%
  Final Inventory     : -56.50 units
  Total Trades        : 147

======================================================================
FINAL RANKING: PREFERRED -> WORST (by Sharpe Ratio)
======================================================================
1. Trending Up              | Sharpe: 4.6939 | MaxDD: 59.41% | PnL: $37731028.76 | Fill: 100.0%
2. Calm Market              | Sharpe: 1.6114 | MaxDD:  4.50% | PnL: $1475.72 | Fill: 100.0%
3. Mean-Reverting           | Sharpe: 1.0014 | MaxDD: 55.56% | PnL: $5872.10 | Fill: 100.0%
4. High Volatility          | Sharpe: 0.9372 | MaxDD: 86.81% | PnL: $-3629.61 | Fill: 100.0%
5. Flash Crash              | Sharpe: 0.5183 | MaxDD: 56.59% | PnL: $-1265.91 | Fill: 100.0%

Simulation complete. All outputs printed above.
```

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